Jake Kressley

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Education

University of Pittsburgh Expected December 2025
Bachelor of Science - Computer Science *GPA: 3.68/4.0*

Relevant Courses
Database Management Systems, Software Quality Assurance, Software Engineering, Artificial Intelligence, Programming Languages for Web Applications, Data Structures & Algorithms, Linear Algebra

Experience

BNY Mellon June 2025 – August 2025
Software Engineer Intern *Lake Mary, FL*

- Constructed a network of RAG AI agents, coordinating agent communication with Python to classify legal documents and extract investor and trade data with 90% accuracy, lowering document review time and boosting operational efficiency by 40%.
- Designed a REST API with Spring Boot to integrate the AI system into an internal Angular-based application, enabling real-time delivery of structured JSON data and improving access to key insights for 30+ internal users.
- Built an automated fund name matching function using Levenshtein distance to map AI-extracted fund names to fund IDs in a 10,000+-row Excel dataset, eliminating manual lookups and reducing processing time by over 50%.

Advantexe Learning Solutions June 2024 – August 2024, June 2023 – August 2023
Web Application Developer Intern *Conshohocken, PA*

- Engineered key components of business acumen simulations, including interactive dashboards and decision-making workflows, contributing to simulation design and delivering React and Bootstrap interfaces that enhanced participant engagement and learning outcomes.
- Researched and recommended the optimal charting API for simulations, cutting implementation time by 30% and improving dashboard performance, enabling participants to access actionable insights for informed decisions.
- Validated and fine-tuned 10+ existing business acumen simulations through black-box testing, ensuring accuracy and reliability to provide participants with a consistent, high-quality learning experience.

Projects

LeetLearn AI | *TypeScript, Angular, Kotlin, Spring Boot, PostgreSQL, OpenAI API, Docker, Google Cloud*

- Developed a scalable backend with Kotlin (Spring Boot), integrating the OpenAI API to generate adaptive multiple-choice quizzes for 200+ LeetCode problems.
- Optimized PostgreSQL schemas with PL/pgSQL functions and materialized views, reducing query latency by 50% and enabling analytics on user accuracy, topic difficulty, and progress trends across quiz attempts.
- Built a full-stack Angular application and containerized it with Docker to ensure consistent deployment, delivering LLM-generated quizzes and explanations for core algorithms and data structures.

Reel Hot Takes | *TypeScript, React, Tailwind CSS, FastAPI, MongoDB, BeautifulSoup, Docker, Google Cloud*

- Built a REST API with FastAPI + MongoDB to ingest Letterboxd user ratings; added a first-page fingerprint (CRC32 of title/rating pairs) that reduced redundant scrapes by 80%.
- Applied machine learning (TF-IDF + cosine similarity) to a 45,000+ film catalog to compute a popularity-weighted "hotness" metric (user vs. global average) and deliver personalized movie recommendations.
- Built a responsive React + TypeScript UI with Tailwind CSS, providing a clean, grid-based layout that scales from mobile to desktop.

Skills

Languages: Python, Java, JavaScript/TypeScript, HTML/CSS, Kotlin

Frameworks: React, Angular, Next.js, Node.js, Express, Tailwind CSS, Spring Boot, FastAPI

Tools/Technologies: REST APIs, Docker, MongoDB, Git/GitHub, Linux, Google Cloud, PostgreSQL